Terry Yu

669-281-6222 | terryyu05@gmail.com | linkedin.com/in/terryyu05 | ty1649.github.io

Mechanical engineering student who enjoys solving interdisciplinary hardware problems from concept through testing.

Education

University of California, Los Angeles — Expected June 2027
B.S. in Mechanical Engineering — GPA: 3.70

Experience

SpaceX — Starbase, TX / Hawthorne, CA
Mechanical Engineering Intern (Starship Crew Mechanisms) — *May 2026 – Present*

- Owned the design and structural analysis of custom fluid-distribution hardware for a new 6-DOF Starship reaction control system, supporting propellant-transfer operations.
- Developed and released 8 custom fittings, manifolds, and sheet-metal weldments through detailed design and analysis, working with suppliers and technicians to resolve manufacturability and integration issues.
- Built an ANSA/Abaqus workflow to evaluate sheet-metal brackets and welds across multiple load cases, documenting margins for design and hardware-readiness reviews.
- Worked across fluids, structures, integration, and manufacturing teams to resolve interface conflicts and adapt hardware to evolving vehicle layouts.

Mechanical Engineering Intern (Falcon Propulsion Components) — *Sep 2025 – Dec 2025*

- Reduced relief-valve cracking pressure by 30% while meeting leakage, vibration, thermal, and pressurization requirements, leading to implementation on new Falcon builds and retrofits across the existing fleet.
- Resolved the redesigned valve's initial vibration-qualification failure by creating and characterizing spring variants, guiding the configuration used for final qualification.
- Created test plans for 11 flight-critical valves with 30 flights of reuse, analyzing pressure, thermal, and vibration data to assess hardware health and support reuse decisions.

DropletPharma Inc. — Los Angeles, CA
Product R&D Intern — *Jun 2025 – Sep 2025*

- Prototyped and tested 10+ dispensing-nozzle iterations for a compact radionuclide-production device, achieving under 5% variation across 100+ tests and supporting medium-scale production.
- Designed a compact fume-extraction assembly within a 10 × 20 × 10 mm envelope, selecting PEEK and PTFE to enable safe chemical handling in a space-constrained device.

Bruin Formula Racing | Project Link — Los Angeles, CA
Drivetrain Lead — *Sep 2023 – Jun 2026*

- Led a 20-member team in developing an electric Formula SAE drivetrain, from concept through competition.
- Developed new torque-transfer and chain-tensioning systems, using topology optimization and nonlinear FEA to reduce drivetrain mass by 25% and rotational inertia by 37%.
- Validated drivetrain design through component and vehicle testing, using results to improve wear resistance, fit, and durability at critical interfaces.

Personal Projects

PIGLR – Paired Outdoor Locator | Project Link — Mar 2026 – Present

- Designed a pocket-sized paired locator device, integrating GNSS positioning, cellular communication, haptic feedback, and embedded electronics.
- Developed a 4-layer PCB and enclosure architecture around antenna placement, component clearances, sensor noise, power, and user-interface constraints.

Technical Skills

CAD & Design: Siemens NX, SolidWorks, GD&T, tolerance analysis, DFM/DFA, engineering drawings
Analysis & Test: ANSYS Mechanical, Abaqus, ANSA, Altair Inspire, vibration and pressure testing
Manufacturing: CNC machining, sheet metal, FDM/SLA prototyping, laser/waterjet cutting
Electronics & Programming: KiCad, C/C++, Python, MATLAB, Zephyr RTOS, BLE, LTE-M, GNSS